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Predictive Model for Silica Nano-Dispersion Gelation to Enhance Conformance Control in Oilfields

E. Basilio and S. Zougheib, Physical Science and Engineering, King Abdullah University of Science and Technology, Saudi Arabia; Q. Sahu, A. Almohsin, and M. Alabdrabalnabi, EXPEC Advanced Research Center, Aramco, Dhahran, Saudi Arabia; H. Hoteit, Physical Science and Engineering, King Abdullah University of Science and Technology, Saudi Arabia

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Abstract

Excessive water production leads to increased treatment costs, elevated carbon emissions, and diminished operational efficiency. To mitigate these adverse effects, this study investigates a novel conformance control method employing a nanofluid system. This innovative approach is designed to effectively intervene in high-permeability zones, wormholes, and fractured reservoirs, thereby optimizing water management practices. Nanofluids incorporating colloidal silica dispersions were selected for their enhanced reservoir penetration capabilities, attributed to their high silica concentration and low viscosity. To elucidate the gelation process of the precursor solution activated by sodium silicate, rigorous rheological characterization was undertaken under diverse conditions, including activator concentration, temperature, water dilution, metal ion presence, and applied shear rate. The resulting data facilitated the development of an analytical model to predict viscosity evolution over time. This model was subsequently validated through a 1D flow experiment in a laboratory-simulated porous medium and subsequently applied to an upscaled numerical simulation of a reservoir featuring a high-permeability layer. A critical parameter, termed the "high viscosity threshold time" (HVTT), was defined to characterize the transition point at which the precursor solution attains a high viscosity state. Kinetic analysis revealed a strong Arrhenius-type dependence of gelation on temperature. To account for the influence of activator concentration and precursor dilution, a modified Arrhenius equation was proposed, providing a robust predictive tool for HVTT under diverse treatment conditions. Furthermore, an exponential model was developed to approximate viscosity evolution over time for specified target viscosity and HVTT values. By integrating these parameters, the model can effectively forecast viscosity evolution throughout the treatment process, offering a predictive framework for the rheological behavior of the system. The numerical simulations successfully validated the analytical model, solidifying its applicability in reservoir scenarios and demonstrating its potential for predictive utilization and optimization in field applications. The integration of this predictive model into field deployment workflows is paramount. Such integration facilitates in-silico evaluation of conformance control strategies, enabling the optimization of treatment efficacy under a wide range of downhole conditions. The synergistic

combination of experimental and numerical methodologies presents a novel approach for optimizing and predicting the success of conformance control treatments in future field applications.

Introduction

The ultimate success of hydrocarbon extraction from subsurface reservoirs is fundamentally governed by two interdependent parameters: displacement efficiency and sweep efficiency. Reservoir conformance, a critical consideration in reservoir management, describes the uniformity with which an injected or mobilizing fluid traverses the reservoir volume under different recovery processes. These processes include natural depletion, water influx drive, secondary water injection, and secondary gas injection. In practice, perfect conformance is rarely achieved due to geological heterogeneities. Features such as high-permeability streaks, thief zones, faults, or fractures can lead to preferential fluid flow pathways that bypass substantial hydrocarbon-rich regions. Such heterogeneities can cause early breakthrough of injected fluids reducing hydrocarbon displacement efficiency and accelerating production decline. Chemical conformance control technologies have been developed to mitigate these challenges. Among these, sodium silicate gels have long been deployed in oil and gas reservoirs, with extensive evidence of their efficacy at field scale (Vinot et al. 1989). These gels are designed to meet two primary operational requirements: (1) low initial viscosity during injection, allowing deep penetration into the reservoir, and (2) delayed gelation to enable in situ viscosity buildup that restricts flow in high-permeability zones, thereby diverting displacement fluids toward less permeable, oil-rich regions.

Colloidal silica nanoparticle dispersions have attracted considerable attention as an alternative to traditional sodium silicate gels. The advantages of colloidal silica originate from intrinsic compositional and rheological differences. Whereas sodium silicate formulations typically possess a relatively low $\text{SiO}_2/\text{Na}_2\text{O}$ ratio (~ 3.4), colloidal silica dispersions often exceed a ratio of 50. This substantially higher silica nanoparticle content can produce stronger, more mechanically robust gels. Additionally, colloidal silica exhibits a viscosity comparable to that of water, in contrast to the syrup-like consistency of sodium silicate solutions, thus enhancing injectability and enabling deeper reservoir penetration. Under suitable activation conditions, colloidal silica dispersions undergo a sol-to-gel transition, forming a three-dimensional particle network tailored to achieve a specific conformance profile (Iler, 1979). This transformation can be modulated by controlling several key parameters such as silica content, which directly affects gel strength and network density; temperature, with elevated temperatures accelerating gelation kinetics; salt concentration, where moderate salinity can promote gelation, but excessive ionic strength may inhibit it; and pH, which governs nanoparticle reactivity and network formation dynamics. A gelling agent, or activator, is typically employed to manipulate these parameters, allowing precise tailoring of gelation behavior for specific reservoir conditions. Prior research has primarily examined salt-based activator systems—including potassium chloride (KCl), sodium chloride (NaCl), and calcium chloride (CaCl_2)—often in conjunction with pH adjustment via hydrochloric acid (HCl) or sodium hydroxide (NaOH) as described in the literature (Alabdrabalnabi et al. 2021; Almohsin et al. 2017; Boul et al. 2015).

The present study focuses on colloidal silica nanoparticle dispersions activated with sodium silicate solution, investigating how operational variables influence gelation kinetics. Specifically, this work examines: (1) the effect of sodium silicate concentration (25–40 wt.%) on silica content and pH; (2) temperature dependence of gelation (75–90 °C) and its correlation with an Arrhenius-type model; (3) the impact of water dilution (0–40 v/v%) on gelation time, and (4) the influence of ionic strength through NaCl addition (0–1 M). A predictive model was developed to estimate gelation time as a function of activator-to-precursor ratio, temperature, and dilution ratio. This model can be integrated into reservoir simulators to forecast the in situ rheological evolution of colloidal silica gels in porous media, with significant implications for improving reservoir conformance, enhancing hydrocarbon recovery, and delaying water breakthrough.

Methodology

Materials

Colloidal silica dispersions are stable aqueous suspensions of amorphous SiO₂ nanoparticles, typically spherical or slightly irregular in morphology. Particle size is influenced by synthesis conditions, particularly temperature (Iler 1979). These dispersions exhibit low viscosity and fluidity, and their optical properties—such as translucence or opalescence—depend strongly on particle size distribution. Gelation of colloidal silica occurs via interparticle cross-linking, whereby discrete nanoparticles form branched aggregates through surface bonding. These aggregates subsequently assemble into a continuous three-dimensional network. The process causes a substantial viscosity increase, ultimately producing a solid-like gel in which the aqueous phase is immobilized by capillary forces within the pore space of the particle network. Gelation does not involve macroscopic changes in silica concentration; rather, it is the spatial organization of particles into a coherent network that governs the transition.

In this work, rheological analysis was employed to quantify the effects of precursor–activator ratio, water dilution, ionic strength, and temperature on gelation kinetics. The experimental setup allowed precise control over each variable while continuously monitoring viscosity development, enabling the identification of gelation onset and progression under various reservoir-relevant conditions.

Experimental Setup

Rheological measurements were conducted using an Anton Paar MCR 302 rheometer equipped with a Peltier temperature-control unit and concentric cylinder geometry (sample volume ~19 mL). The system allowed precise temperature control (± 0.2 °C) over a range of 0–200 °C. Most measurements were performed at a shear rate of 5 s^{-1} , representative of average reservoir conditions.

Experimental Procedure

The experimental workflow comprised two sequential stages: (i) preliminary static gelation screening to identify viable formulation parameters and (ii) dynamic rheological measurements to monitor gelation kinetics. Prior to rheological testing, static gelation experiments were conducted to define parameter ranges that would yield gel formation within a practical operational timescale. These trials investigated the influence of activator concentration, water dilution, and brine dilution on gelation behavior. For each formulation, the precursor component was first weighed to the target weight percentage (wt.%) specified for the system. The activator was subsequently introduced at the predetermined wt.%. In dilution experiments, the homogenized solution was further diluted with deionized water (v/v%) or brine to the desired ratio. The brine was prepared by dissolving sodium chloride (NaCl) in deionized water to a salinity of 50,000 ppm. Following preparation, each formulation was transferred into sealed containers and stored at 90 °C—selected as the reference temperature for screening, corresponding to the highest operational temperature used in later rheological tests. This choice reflects the well-established inverse relationship between gelation rate and temperature, whereby lower temperatures typically slow network formation. To minimize evaporation, a thin protective oil layer was placed over the sample surface. The gelation endpoint was defined as the transition from a freely flowing fluid to a self-supporting gel within a 20-hour observation period. Activator concentrations of 25 wt.% or higher induced gelation, whereas the 20 wt.% formulation exhibited no appreciable viscosity increase. Regarding the effect of water dilution, gelation occurred only when dilution ratios were below 50%. On the other hand, brine dilution (0–90 v/v%) in the same 35 wt.% activator system produced gelation across all samples, although high brine fractions (80% and 90%) led to observable aggregation during gel formation. Formulations identified as viable from the static tests were subsequently subjected to rheological characterization. The prepared gel system was transferred into the rheometer's measurement cup, and the system temperature was ramped from 20 °C to the specified test temperature. Viscosity and other rheological parameters were continuously recorded at 5-second intervals

using the instrument's data acquisition software. These measurements enabled real-time tracking of gelation kinetics and facilitated quantitative analysis of the influence of formulation and environmental variables on the gelation process.

Experimental Results

A parametric investigation was conducted to quantify the influence of selected formulation and environmental variables on colloidal silica gelation kinetics. The experimental design incorporated four primary factors: (i) precursor–activator concentration, (ii) water dilution ratio, (iii) temperature, and (iv) electrolyte concentration. The precursor–activator concentration was adjusted to four discrete levels of 25, 30, 35, and 40 wt.% to evaluate compositional effects. Each concentration was subjected to rheological testing at four target temperatures (75, 80, 85, and 90 °C) in order to capture the coupled influence of chemical formulation and thermal conditions. The role of aqueous dilution was examined by preparing silicate gel systems with activator concentrations of 30, 35, and 40 wt.% and blending them with deionized water at volumetric dilution ratios from 0 to 40 v/v%. All water-diluted systems were assessed at 90 °C. Electrolyte effects were evaluated by introducing sodium chloride (NaCl) at concentrations ranging from 0 to 1 M into a 35 wt.% activator system at a fixed dilution ratio of 20 v/v%. These electrolyte-containing formulations were likewise tested at 90 °C. In all experimental cases, the gelation endpoint was defined as the moment when the viscosity profile exhibited a distinct and rapid escalation, indicating the transition from a low-viscosity sol to a high-viscosity gel network.

The hydrogen ion concentration of the gel system exerts a decisive influence on gelation kinetics by modulating the electrostatic characteristics of silica nanoparticle surfaces. These surface charges govern the magnitude and nature of interparticle interactions, thereby affecting the probability of particle collisions and the formation of stable interparticle bonds—processes that are fundamental to the development of a continuous gel network. Under more acidic conditions, where hydroxide ion (OH^-) concentration is reduced, attractive forces between particles become dominant, enhancing collision frequency and promoting the onset of gelation within silicate-based systems. In parallel, the abundance of silicate species within the formulation also plays a crucial role. A higher silicate content accelerates gel formation by increasing the availability of structural units necessary for constructing the three-dimensional particle network. In the present work, a sodium silicate solution served as the activator. Upon introduction into the colloidal silica precursor, the activator simultaneously lowers the system's pH and augments its silicate concentration. This coupled effect enhances the rate of gelation by both strengthening interparticle attraction and increasing the supply of network-forming species.

For the purpose of quantitatively identifying the onset of the rapid gelation phase, the parameter High-Viscosity Threshold Time (HVTT) was employed. In the present study, HVTT is defined as the elapsed time required for a gel formulation to attain a viscosity of 50 mPa·s, representing a transition point beyond which the gelation rate exhibits a pronounced acceleration. To assess the sensitivity of this metric, additional evaluations were conducted using an alternative threshold value of 300 mPa·s, enabling comparison of gelation trends under different viscosity criteria.

Temperature Concentration Relationship

Elevated temperatures enhance the kinetic energy of silica nanoparticles, intensifying Brownian motion and increasing particle mobility. This heightened dynamic activity leads to a greater frequency of interparticle encounters, thereby improving the probability that collisions will surmount the activation energy barrier necessary for the establishment of stable particle–particle bonds. As a consequence, gel network formation is accelerated at higher temperatures. Activator concentration is another key determinant of gelation rate, primarily through its effect on the total silicate content available for network assembly. Higher concentrations supply a greater number of structural precursors, expediting the formation of the

three-dimensional gel framework. Figure 1 summarizes the High-Viscosity Threshold Time (HVTT) for all tested temperature–activator combinations. The results show that HVTT decreases with increasing activator concentration, and for a fixed concentration, higher temperatures also yield shorter HVTT values, confirming the synergistic effect of these two parameters on gelation rate.

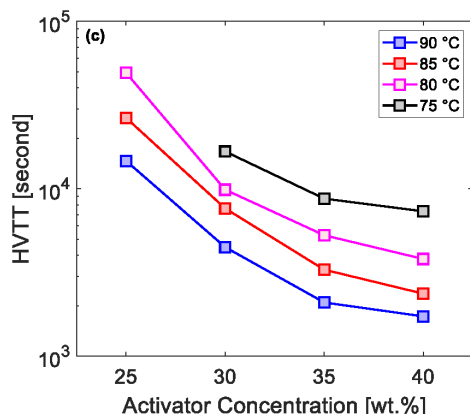


Figure 1—Corresponding High-Viscosity Threshold Time (HVTT) values determined for each temperature–activator concentration combination.

Water Dilution Effects

Dilution of silicate gel systems with deionized water was found to significantly retard the gelation process. This behavior is primarily attributable to the simultaneous reduction in precursor and activator concentrations, which decreases the overall population of silica particles and thereby reduces the frequency of interparticle collisions. A lower particle collision rate slows the progression toward network formation. Furthermore, increased dilution expands the mean separation distance between individual particles, diminishing attractive forces and impeding the establishment of a continuous gel framework. At sufficiently high dilution levels, the silica particle density may fall below the percolation threshold required for three-dimensional network formation, effectively preventing gelation. Even when gelation does occur under diluted conditions, the resulting networks are expected to be mechanically weaker and characterized by larger pore structures relative to gels formed at higher silica concentrations. The relationship between dilution and gelation rate is presented in Figure 2, which shows the variation in HVTT for dilution ratios of 0, 10, 20, and 30 v/v% at fixed activator concentrations of 30, 35, and 40 wt.% and a constant temperature of 90 °C.

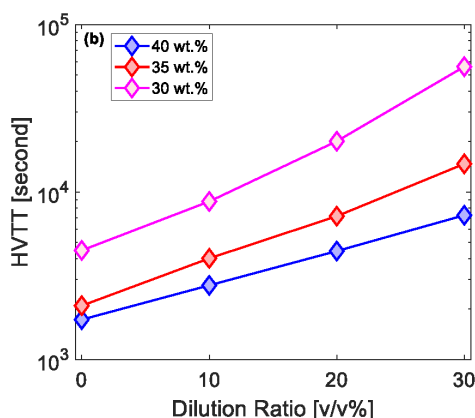


Figure 2—High viscosity threshold time (HVTT) at fixed activator concentrations (30, 35, 40 wt.%).

Electrolyte Effects

The gelation kinetics of colloidal silica systems are strongly affected by both the concentration and the chemical nature of dissolved electrolyte species. These effects arise primarily from their influence on surface charge interactions, reaction rates, and particle aggregation dynamics. Cations introduced via electrolyte solutions interact with the negatively charged silanol groups present on silica particle surfaces, thereby reducing the electrostatic repulsion that normally inhibits close particle approach. This reduction in repulsive forces enhances the probability of effective particle–particle contacts, enabling attractive interactions to dominate and facilitating the development of a three-dimensional gel network. In addition to modifying interparticle forces, the presence of electrolytes elevates the ionic strength of the medium, which can promote the hydrolysis and condensation of silanol groups through catalytic action by the dissolved metal ions (Quarch and Kind 2010; Hamouda and Amiri 2014). These catalytic effects accelerate the bond formation process within the gel matrix, leading to a faster transition from a dispersed sol to a solid-like gel structure. Sodium chloride (NaCl) solutions were examined at concentrations of 0, 0.25, 0.50, 0.75, and 1 M. Gel systems were prepared by first formulating a base silicate gel with an activator concentration of 35 wt.% and subsequently mixing it with brine at a volumetric ratio of 20 v/v%. The quantitative effect of electrolyte concentration on gelation rate is illustrated in Figure 3, where progressively higher NaCl concentrations correspond to reduced HVTT values.

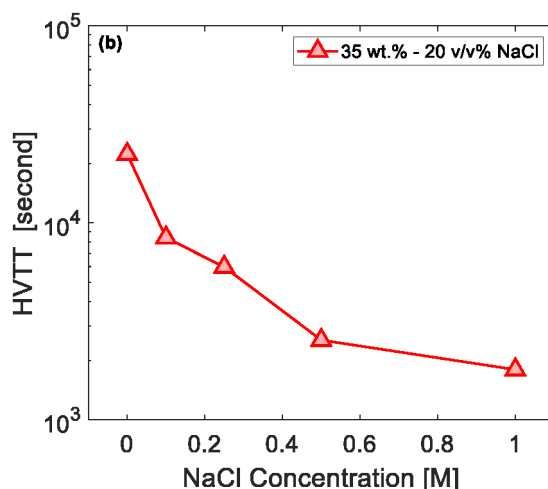


Figure 3—High viscosity threshold time (HVTT) at fixed electrolyte concentrations (NaCl 0, 0.10, 0.25, 0.50, 1.0 M) at a 20 v/v% dilution ratio for a 35 wt.% activator concentration gel system.

Analytical Model Development

The influence of temperature on gelation kinetics has been widely documented, with numerous studies reporting that, under constant pH and ionic strength, the process frequently adheres to a first-order Arrhenius-type relationship (Jurinak and Summers 1991). This formulation expresses the dependence of the reaction rate constant (k) on the absolute temperature (T) through an exponential function. Within the context of gelation, an increase in the reaction rate constant (k) reflects an accelerated transition from sol to gel, thereby reducing the gelation time. Consequently, the High-Viscosity Threshold Time (HVTT), as previously defined, exhibits an inverse relationship with k as presented in Equation 1, where A is the pre-exponential factor related to the gelation kinetics.

$$HVTT = Ae^{\frac{E_a}{RT}} \quad (1)$$

Although the Arrhenius equation expresses the temperature dependence of reaction rates primarily through its exponential component, the pre-exponential factor (A) embodies additional influences, including the effect of reactant concentration. This parameter is often associated with the reaction order, reflecting the

number of reactant species participating in the rate-limiting step. The standard Arrhenius formulation does not explicitly incorporate concentration terms. The influence of concentration on A indicates that increasing the activator concentration leads to a proportional rise in $\ln(A)$, implying a higher frequency of successful molecular collisions. Such behavior can be attributed to the role of the activator in promoting more effective particle interactions, thereby enhancing collision efficiency and accelerating the overall gelation reaction.

The High-Viscosity Threshold Time (HVTT) model, initially expressed as a function of temperature and activator concentration, can be extended to incorporate the influence of water dilution. This refinement involves calculating the modified pre-exponential factor, $\ln(A')$, for gel systems prepared at different dilution ratios and applying multilinear regression to relate $\ln(A')$ and the dilution ratio to HVTT. Through this statistical approach, a unified expression is derived that simultaneously accounts for activator concentration, temperature, and water dilution. The resulting formulation, presented in Equation 2, provides a more robust predictive framework for estimating HVTT by integrating the combined effects of these three key variables.

$$\ln(HVTT) = 12595 \times [1 + v/v] \left(\frac{1}{T} \right) + 0.0109 \times \left(\frac{1}{C^4} \right) - 29.215 \times [v/v] - 27.763 \quad (1.2)$$

Conclusions

The present study introduces a predictive model that demonstrates high accuracy in estimating the High-Viscosity Threshold Time (HVTT) for colloidal silica dispersions within the experimentally validated parameter space: temperatures between 75 °C and 90 °C, activator concentrations ranging from 25 wt.% to 40 wt.%, and water dilution ratios between 0 v/v% and 30 v/v%. Although application outside these ranges may necessitate recalibration of the model parameters, the strong agreement observed between experimental measurements and predicted values highlights the robustness of the approach.

By explicitly incorporating the combined effects of temperature, silica concentration, and dilution ratio, the model provides a comprehensive framework for anticipating gelation behavior under varying operational conditions. This capability positions the model as a valuable design tool for the engineering of layered hydraulic barriers in highly permeable formations. In practice, it enables the determination of an optimal precursor-to-activator ratio to achieve a target gelation time, thereby controlling the penetration depth of the treatment fluid. Such predictive capability supports the formulation of customized treatment strategies, ensuring that barrier design is adapted to site-specific requirements and performance objectives, ultimately improving the operational efficiency and success of colloidal silica gel applications. Future work includes assessing mechanical properties such as elasticity and compressive resistance.

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